Health Deterioration Detection in Hypertensives

Sponsors

1. Sentinel Healthcare
2. University of Washington, MSDS

Team Members

1. Agustin Lew (ablew@uw.edu)
2. Ankit Tandon (antand@uw.edu)
3. Patrick King (pking70@uw.edu)
4. Suman Bhagavathula (sumanbh@uw.edu)

Abstract

Predicting the gradual unraveling of cardiac protection in hypertensive patients is a major challenge for clinicians, who rely on limited data collected sporadically during scheduled patient visits. Sentinel Healthcare has developed a Remote Physiologic Monitoring platform to gather and analyze remotely collected patient data. Patients use provided blood pressure cuffs to take their own vital signs regularly and upload the data to Sentinel's app platform. Healthcare providers can use this data to better provide care and intervention for at-risk patients. This project reports preliminary results on modeling the collected blood pressure and heart rate data to better predict if a patient might experience a health deteriorating event in the near future.

Modelling has been based on three different techniques: Long Short Term Memory (LSTM), Vector AutoRegression (VAR) and Seasonal AutoRegressive Moving Average (SARIMA). We found that LSTM performs best in predicting SBP while VAR performs best in predicting HR. We believe this is because VAR considers the interaction effects between HR and SBP and SBP is harder to predict given HR, while the opposite is not true according to Johansen's cointegration test. We also produced an algorithm that raises flags when the model suggests that health condition will deteriorate. This algorithm is based on multiple different conditions, such as high blood pressure or rising heart rate. We found that VAR performs best in finding health deterioration signals.

The models show potential for time series modeling on vitals data. Initial results were promising, and we think these models are a framework to which additional signals such as Beta Blocker dosages can be added for robustness.